

Matteo Robbiati

Ph.D. Candidate in Physics

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Summary

Ph.D. candidate in quantum computing and machine learning at the European Organization for Nuclear Research (CERN), founded by the Quantum Technology Initiative and supervised by the theoretical physics group at University of Milan. During my Ph.D., I am working on quantum machine learning algorithms applied to integration, error mitigation, multi-dimensional regression and ground state preparation. I am contributing to the development and maintenance of open-source, full-stack frameworks for quantum computing as part of the Qibo collaboration. In the same context, I am managing and developing educational materials to make our open-source platform more accessible to users of all skill levels, from beginners to advanced.

Experience

- 2022 - present** **Doctoral researcher at CERN**, funded by the CERN's Quantum Technology Initiative and supervised by the theoretical physics group of the University of Milan.
- 2016 - 2022** **Private teaching** of maths and physics to high school students.
- 2021** **Research and data analysis for the cooperative “La Valle di Ezechiele”**, we investigated how work outside prison can help reduce prisoners' re-offending rate. We also examined case studies on the use of social impact bonds to support employment opportunities for ex-prisoners.

Education

- 2022 - present** **Ph.D. in Physics**, European Organization for Nuclear Research (CERN), Switzerland.
- 2019 - 2022** **Master degree in Physics**, University of Milan, Italy.
- 2015 - 2019** **Bachelor degree in Physics**, University of Milan, Italy.

Skills

- **Programming:** Python, C/C++, HTML, Nextjs, BASH, Javascript.
- **Framework & Libraries:** NumPy, SymPy, SciPy, Keras, Pytorch, Tensorflow, Scikit-learn, Jax, Pandas, Qibo, Qiskit, Cirq, PennyLane, MPI.
- **Graphic & Productivity:** Matplotlib, Seaborn, Plotly, Inkscape, Xmind, Canva, Microsoft Office, Krita, GIMP.
- **Operating System:** Linux, Microsoft, MacOS.
- **Tools:** git, tmux, slurm.
- **Languages:** Italian (native), English (fluent), Spanish and French (intermediate).

Software projects

Below, the first list includes packages that are publicly distributed and regularly maintained, the second contains code developed for specific projects that is not maintained, and the third refers to collections of graphics and presentations. All the listed projects are open-source and can be found at: <https://github.com/MatteoRobbiati>.

Community-driven packages

- **Qibo** - A full-stack framework for quantum computing ([Core developer](#));
- **Qiboml** - A Full-stack hybrid classical-quantum machine learning framework ([Core developer](#));
- **Qiboedu** - Quantum computing educational activities using Qibo ([Core developer](#));
- **Qibolab** - Quantum hardware module and drivers for Qibo ([Contributor](#));
- **Qibocal** - Quantum calibration, characterization and validation module for Qibo ([Contributor](#)).

Project-specific tools

- **QiNNtegrate** - Multi-variable numerical integration via Quantum Machine Learning ([Core developer](#));
- **adiabatic-fit** - Probability density estimation via hybrid gate&analog-based quantum computing ([Core developer](#));
- **rtqem** - Real-time quantum error mitigation for variational optimization on quantum hardware ([Core developer](#));
- **boostvqe** - Boosting ground-states preparation with Double-Bracket Quantum Algorithms ([Core developer](#)).

Graphic and presentation

- **LaTeX-files** - Collection of documents written with LaTeX;
- **inkscape-drawings** - Collection of scalable vector graphics produced with Inkscape;
- **xmind-diagrams** - Collection of diagrams produced with Xmind.

Publications

Peer-reviewed

- 2025** Simulating Bell inequalities with Qibo, *I. Masina, G. Lo Presti, [M. Robbiati](#), M. Grossi*, [arXiv:2501.01434](#)
- 2025** Determining probability density functions with adiabatic quantum computing, *[M. Robbiati](#), J. M. Cruz-Martinez, S. Carrazza*, [Quantum Mach. Intell. Vol. 7, art. 5](#)
- 2024** Qibolab: an open-source hybrid quantum operating system, *S. Efthymiou, A. Orgaz-Fuertes, R. Carobene, J. Cereijo, A. Pasquale, S. Ramos-Calderer, S. Bordoni, D. Fuentes-Ruiz, A. Candido, E. Pedicillo, [M. Robbiati](#), Y. Paul Tan, J. Wilkens, I. Roth, J. I. Latorre, and S. Carrazza*, [Quantum 8, 1247](#)
- 2024** Multi-variable integration with a variational quantum circuit, *J. M. Cruz-Martinez, [M. Robbiati](#) and S. Carrazza*, [Quantum Sci. Technol. 9, 035053](#)
- 2024** Characterization of a Transmon Qubit in a 3D Cavity for Quantum Machine Learning and Photon Counting, *A. D'Elia, B. Alfakes, A. Alkhazaleh, L. Banchi, M. Beretta, S. Carrazza, F. Chiarello, D. Di Gioacchino, A. Giachero, F. Henrich, A. S. Piedjou Komnang, C. Ligi, G. Maccarrone, M. Macucci, E. Palumbo, A. Pasquale, L. Piersanti, F. Ravaux, A. Rettaroli, [M. Robbiati](#), S. Tocci and C. Gatti*, [Appl. Sci. 14\(4\), 1478](#)

Pre-prints

- 2024** Qibocal: an open-source framework for calibration of self-hosted quantum devices, *A. Pasquale, E. Pedicillo, Juan Cereijo, S. Ramos-Calderer, A. Candido, G. Palazzo, R. Carobene, M. Gobbo, S. Efthymiou, Y. Paul Tan, I. Roth, [M. Robbiati](#), J. Wilkens, A. Orgaz-Fuertes, D. Fuentes-Ruiz, A. Giachero, F. Brito, J. Ignacio Latorre, S. Carrazza*, [arXiv:2410.00101](#)
- 2024** Double-bracket quantum algorithms for high-fidelity ground state preparation, *[M. Robbiati](#), E. Pedicillo, A. Pasquale, X. Li, A. Wright, R. M. S. Farias, K. Uyen Giang, J. Son, J. Knörzer, S. Thye Goh, J. Yong Khoo, N. H.Y. Ng, Z. Holmes, S. Carrazza, M. Gluza*, [arXiv:2408.03987](#)
- 2023** Real-time error mitigation for variational optimization on quantum hardware, *[M. Robbiati](#), A. Sopena, A. Papaluca, S. Carrazza*, [arXiv:2311.05680](#)

Proceedings of Science

- 2024** An open-source framework for quantum hardware control, *E. Pedicillo, A. Candido, S. Efthymiou, H. Sargsyan, Y. Paul Tan, J. Cereijo, J. Yong Khoo, A. Pasquale, [M. Robbiati](#), S. Carrazza*, [arXiv:2407.21737](#)
- 2024** Beyond full statevector simulation with Qibo, *A. Pasquale, A. Papaluca, R. M. S. Farias, [M. Robbiati](#), E. Pedicillo, S. Carrazza*, [arXiv:2408.00384](#)
- 2024** Strategies for optimizing double-bracket quantum algorithms, *L. Xiaoyue, [M. Robbiati](#), A. Pasquale, E. Pedicillo, A. Wright, S. Carrazza, M. Gluza*, [arXiv:2408.07431](#)
- 2023** A quantum analytical Adam descent through parameter shift rule using Qibo, *[M. Robbiati](#), S. Efthymiou, A. Pasquale, S. Carrazza*, [Volume 414 - 41st International Conference on High Energy Physics](#)

Participation in research grants

2021 - 2022 Collaborator, Automatic Monte Carlo on GPU, *Linea 2A, University of Milan, Italy*.

Contributions in conferences, schools and workshops

- 2025** Double-bracket quantum algorithms for high-fidelity ground state preparation, *QT4HEP, CERN*
2024 Quantum Machine Learning in High Energy Physics with Qibo, *PyHEP workshop, Online*
2024 Real-time error mitigation for variational optimization on quantum hardware, *ACAT 2024, Stony Brook University, New York, USA*
2024 Full-stack Quantum Machine Learning in High Energy Physics, *APS March Meeting 2024, Minneapolis, USA*
2024 Full-stack Quantum Machine Learning using Qibo, *Singapore Quantum Tech Meetup, Singapore*
2023 Real-time error mitigation for variational optimization on quantum hardware, *QTML2023, CERN*
2023 Determining probability density functions with adiabatic quantum computing, *QTML2023, CERN*
2023 qibo: a full-stack framework for simulation, control and calibration of self-hosted qubit devices, *Workshop INFN CSN4&5, Torino*
2023 Determining probability density functions with adiabatic quantum computing, *EQAI2023, Udine*
2023 Full-stack Quantum Machine Learning for High Energy Physics, *Milan Christmas Meeting 2023, Milan*
2023 Full-stack Quantum Machine Learning, *Students workshop of Milan University Ph.D. school, Milan*

Attended schools

- 2024** Spring School on Near-Term Quantum Computing, *Benasque, Spain*
2023 European Summer School on Quantum AI, *Udine, Italy*

Teaching and educational contributions

- Online course** Quantum computing with Qibo, *Centre of Quantum Technologies, Singapore*
Course Minicourse on quantum computing, *ICTP, Sao Paulo, Brazil*
Ph.D. school course Quantum Machine Learning with Qibo, *Como, Italy*
Master thesis co-supervision Towards advanced data encoding techniques for quantum re-uploading models, *Niccolò Laurora, University of Milan, Milan, Italy*
Bachelor thesis co-supervision Momentum-Based Real Time Quantum Error Mitigation in Variational Quantum Algorithms, *Alessandra Esmeraldi, University of Milan Bicocca, Italy*
Master thesis co-supervision Tensor Network Born Machine, *Mattia Robbiano, University of Turin, Italy*
Bachelor thesis co-supervision The role of non-linear encodings in quantum data reuploading, *Laura Pedro-Botet Ezquerro, Universitat Politècnica de Catalunya, Spain*

Outreach and volunteering

- 2016 - 2022** **Scout educator in AGESCI**, My role involved organizing and managing educational activities and social impact initiatives in the community approximately twice a week, typically involving between 5 and 30 young people aged 13 to 17.
2019 - 2022 **Volunteer with APWOYO Onlus**, we distributed food and clothes to the homeless in Milan two nights per month.

Reference contacts

- **Stefano Carrazza** - Associate professor at University of Milan - stefano.carrazza@cern.ch
- **Marek Gluza** - Postdoctoral Researcher at NTU, Singapore - marekludwik.gluza@ntu.edu.sg
- **Juan Manuel Cruz Martínez** - Senior research fellow at CERN - juan.manuel.cruz.martinez@cern.ch
- **Yousef Husein** - Senior Nuclear Engineer at Blykalla, Sweden - Yousef.husein@blykalla.com